

INDIRA GANDHI NATIONAL OPEN UNIVERSITY

School of Computer and Information Sciences

Maidan Garhi, New Delhi — 110 068

PROJECT REPORT

BCSP-064

DEVELOPMENT OF AN E-COMMERCE WEB APPLICATION FOR ONLINE SHOPPING

*A project report submitted in partial fulfilment of the requirements
for the award of the degree of*

BACHELOR OF COMPUTER APPLICATIONS (BCA)

Particulars	Details
Submitted by	Kailash Kumar Jha
Enrolment Number	2400767095
Programme	Bachelor of Computer Applications (BCA)
Course Code	BCSP-064
Study Centre Code	07162P
Regional Centre	RCD1 — 07: Delhi 1 (Mohan Estate, South Delhi)
E-mail	Provided in the submitted copy
Mobile Number	Provided in the submitted copy
Project Guide	Shankar Jha — Backend Lead (Software Engineer)
Session	January 2026

Under the supervision of

Shankar Jha

Backend Lead (Software Engineer), Wizcart Technologies LLC

CERTIFICATE OF ORIGINALITY

This is to certify that the project report entitled "DEVELOPMENT OF AN E-COMMERCE WEB APPLICATION FOR ONLINE SHOPPING" submitted to Indira Gandhi National Open University in partial fulfilment of the requirements for the award of the degree of BACHELOR OF COMPUTER APPLICATIONS (BCA), is an original work carried out by Mr. Kailash Kumar Jha, Enrolment No.: 2400767095, under the guidance of Mr. Shankar Jha.

The matter embodied in this project is a genuine work done by the student and has not been submitted whether to this University or to any other University / Institute for the fulfilment of the requirement of any course of study.

Signature of the Student	Signature of the Guide
Name: Kailash Kumar Jha	Name: Shankar Jha
Enrolment No.: 2400767095	Designation: Backend Lead (Software Engineer)
Address: Provided in the submitted copy	Address: Provided in the submitted copy
Date:	Date:

SCHOOL OF COMPUTER AND INFORMATION SCIENCES

IGNOU, MAIDAN GARHI, NEW DELHI — 110 068

PROFORMA OF BCA PROJECT PROPOSAL (BCSP-064)

(Project Title and Guide's Details)

#	Item	Details
	Enrolment No.	2400767095
	Regional Centre Code	RCD1
	Study Centre Code	07162P
1	Name of Student	Kailash Kumar Jha
2	Address of the Student	Provided in the submitted copy
3 (a)	E-mail	Provided in the submitted copy
3 (b)	Telephone / Mobile No.	Provided in the submitted copy
4	Title of the Project	DEVELOPMENT OF AN E-COMMERCE WEB APPLICATION FOR ONLINE SHOPPING
5 (a)	Name of the Project Guide	Shankar Jha
5 (b)	Designation of the Project Guide	Backend Lead (Software Engineer)
6	Address of Project Guide	Provided in the submitted copy
7	Qualification of the Guide	B.Tech (Information Technology) — bio-data attached
8	Industrial / Teaching experience of the Guide	5+ years (Industry)
9	Software used for this Project	MongoDB 7 (Community Server), Express.js 4, React 18, Node.js 22, Mongoose 8, JSON Web Token, bcrypt, Jest, Supertest, Vite, Git, VS Code, Postman

Signature of the Student	Signature of the Guide
Date:	Date:

Note — as required by the BCSP-064 guidelines: the use of Visual Basic with MS-Access as front end and back end respectively is forbidden, and the use of C or C++ is prohibited for projects associated with database management. This project uses none of those. MongoDB is listed among the permitted

database engines and React among the permitted languages and frameworks in Section VII of the guidelines.

For Office Use Only	
Approved ☐ Not approved ☐	Signature, Designation and Stamp of the Project Proposal Evaluator Date:
Suggestions for reformulating the project:	

BIO-DATA OF THE PROJECT GUIDE

Particulars	Details
Name	Shankar Jha
Qualification	B.Tech (Information Technology), Bharati Vidyapeeth College of Engineering
Current Designation	Backend Lead (Software Engineer)
Organisation	Wizcart Technologies LLC
Total Experience	5+ years (Industry)
Area of Specialisation	Backend development, product engineering, SaaS platforms
Address	Provided in the submitted copy
E-mail	Provided in the submitted copy

Professional Summary

Shankar Jha is a software engineer with more than five years of industry experience in backend engineering, full-stack application development and SaaS platform architecture. He currently serves as Backend Lead at Wizcart Technologies LLC, where he is responsible for the design of scalable service infrastructure, data synchronisation pipelines and transactional API design.

Professional Experience

- **Backend Lead** — Wizcart Technologies LLC (current role)
- **Senior Development Engineer** — AVIZVA
- **Full Stack Engineer** — Surepass Technologies
- **Full Stack Software Developer** — Flowboard Technologies

Key Projects and Achievements

- Designed and engineered core systems for high-throughput, enterprise-grade SaaS platforms.
- Architected security-compliance protocols and transactional APIs for financial technology platforms.
- Integrated distributed analytical engines within real-time health-technology systems.

Educational Credentials

- **B.Tech in Information Technology** — Bharati Vidyapeeth College of Engineering

Signature of the Guide	Date:
------------------------	-------------

ACKNOWLEDGEMENT

The completion of this project has depended on the guidance, encouragement and patience of a number of people, and it is a pleasure to record my thanks to them here.

I am deeply grateful to my project guide, Mr. Shankar Jha, Backend Lead at Wizcart Technologies LLC, for agreeing to supervise this work. His insistence that I justify every design decision — rather than simply make it work — shaped the project more than any other single influence. The concurrency handling in the checkout engine and the decision to recompute every price on the server are both the direct result of his review comments.

I thank the School of Computer and Information Sciences, IGNOU, for framing a project course that requires the student to work through the whole software development life cycle rather than only the coding phase. I am also grateful to the Coordinator and the academic counsellors at Study Centre 07162P for their guidance on the procedural requirements of the submission.

Finally, I thank my family for their steady support over the months this project took, and for tolerating the long evenings it consumed.

Kailash Kumar Jha
Enrolment No.: 2400767095

INDIRA GANDHI NATIONAL OPEN UNIVERSITY

MAIDAN GARHI, NEW DELHI — 110068

REMUNERATION BILL FOR THE BCA PROJECT GUIDE

#	Item	Details
1	Course Code	BCA (BCSP-064)
2	Name of the Guide	Shankar Jha
3	Residential Address	Provided in the submitted copy
4	Designation	Backend Lead (Software Engineer)
5	Office Address	Wizcart Technologies LLC

This is to certify that I have guided the following student for their project work:

S. No.	Enrolment Number	PR No. (filled by the Regional Centre)	Name of the Student	Title of the Project	Amount Claimed
1	2400767095		Kailash Kumar Jha	DEVELOPMENT OF AN E-COMMERCE WEB APPLICATION FOR ONLINE SHOPPING	
2					
3					

Signature of the Guide	Date:
------------------------	-------------

NOTE: A project guide cannot guide more than eight students at any given point in time. This form, duly signed by the guide, should be placed in a SEPARATE ENVELOPE and submitted along with the project report. A remuneration bill not accompanying the project report will not be considered for payment.